

Weekly Report: Week 3

2026-08-12

1 Scope

This report covers the pair-adaptive attribute-selection stage (Step 1.5) of the LLM-assisted entity resolution pipeline, measured on Fodors–Zagat and DBLP–ACM: the method, the experimental design the comparisons require, the results, and what those results do and do not establish.

All numbers come from runs recorded on 2026-08-12 and 2026-08-18. The matcher LLM is `deepseek-v4-flash-0731` (served through OpenRouter) and the embedding model is `bge-large-en-v1.5`. Results predating this report are not comparable and are not quoted: they were produced under a different matcher LLM and before several defects in the experimental setup were repaired. Amazon–Walmart is not covered; it is running now and is the subject of the next report.

The headline result is that on DBLP–ACM, showing the matcher a small, pair-dependent subset of attributes beats both showing it everything and showing it a fixed subset chosen with labelled data — but only under a selection rule that can express small subsets, and the rule matters as much as the method does.

2 Pipeline Overview

The pipeline runs in five stages, each timed and recorded in a per-run JSON summary under `logs/<dataset>/runs/`.

Load and normalize. Both tables and the ground truth are read, ID columns are dropped, and every remaining field is normalized: HTML entity references are decoded, text is Unicode-decomposed with combining marks stripped, and address and directional abbreviations are standardized. This matters more than it appears: `ACM.csv` stores 461 HTML entity references (`ö`, `—`) where `DBLP.csv` stores none, so without decoding, the same author name is two unrelated strings across the two sources.

Embed and block. Each attribute column is encoded independently with SentenceBERT, the column vectors are concatenated and L2-normalized into one embedding per record, and multi-probe locality-sensitive hashing retrieves the top- k most similar Table-B candidates for every Table-A record. On DBLP–ACM this reduces 6,001,104 possible pairs to 13,080 candidates while retaining all 2,224 ground-truth matches.

Step 1 — global tuple condensation (optional). One attribute subset is chosen once for the whole dataset and applied to every record. The tables may optionally be re-blocked on the condensed attributes; see Section 4.2.

Step 1.5 — pair-adaptive attribute selection (optional). A small sample of candidate pairs is profiled by the LLM for per-attribute importance, and a hybrid predictor/retriever model transfers a *different* attribute subset to every candidate pair. Blocking is unchanged, so the candidate set is identical to the no-selection baseline.

Match. One LLM call decides Yes/No per candidate pair, using only the attributes selected for that pair.

Metrics and logging. Precision, recall, and F1 are computed end-to-end, against all ground-truth matches, so pairs lost at blocking count against recall.

3 Methodology

3.1 Step 1.5: Pair-Adaptive Attribute Selection

Step 1.5 has a profiling stage and a transfer stage.

Profiling. A label-free sampler selects a small set of candidate pairs (20 on Fodors–Zagat, 50 on DBLP–ACM) and the LLM is asked, for each, which attributes matter for deciding whether the two records describe the same entity. Ground truth is not consulted at any point. The **diversity** sampler is used throughout this report, holding the sampler fixed so that other variables can be read.

Transfer. For every candidate pair, six cheap similarity features are computed per attribute — token overlap, edit similarity, exact match, semantic similarity, numeric similarity, and a numeric-type indicator — without calling the LLM. Two models then predict per-attribute importance from those features: a Ridge regressor fit on the profiled pairs, and a nearest-neighbour retriever that averages the importance vectors of the five most feature-similar profiled pairs. Their outputs are blended equally and normalized to sum to one, and a selection rule decides how many of the ranked attributes to keep.

The profiling stage costs about 1% of a run’s tokens. Its purpose is to buy a rule that can be applied to the remaining 99% without further LLM calls.

3.2 Two Ways to Elicit Importance

The profiling prompt is one of the two variables this report isolates. Two formulations were compared, identical in framing and guardrails and differing only in the answer requested.

Ordinal. The LLM assigns each attribute an integer score from 0 (not useful) to 3 (decisive), scoring each attribute independently.

Decisive. The LLM returns the smallest subset of attributes sufficient to make the decision, weighing attributes against one another in a single judgement.

The motivation for the second formulation was that the recorded 0–3 scores were not being used discriminatively. Aggregated over all runs, DBLP–ACM used score 0 on 1% of judgements, collapsing a four-point scale to three; Fodors–Zagat used score 3 on 55%. An importance vector that is nearly flat cannot express that one attribute matters more than another, whatever selection rule consumes it.

3.3 How Many Attributes to Keep

Profiling and transfer produce a ranked importance vector per pair. A separate rule turns that ranking into a subset, and it is the second variable isolated here.

Cumulative keeps attributes in descending importance until their shares sum past a fixed target, by default 0.8. It is the pipeline’s historical default and is not scale-free: importance is normalized to sum to one, so on a wide schema each attribute’s share shrinks and a fixed target becomes unreachable. On Amazon–Walmart’s 13 attributes the 0.8 target needs 7.7 attributes on

average against a cap of 5, so the cap binds for every pair and the rule cannot express “this pair needs fewer fields” at all.

Gap cuts at the largest drop in the ranked importance profile. It has no threshold to tune and does not saturate on any of the three schemas.

Both rules are subject to a floor and a cap on the number of attributes kept. The floor was 2 by default and is set to 1 wherever a run is allowed to reduce a pair to a single attribute.

4 Experimental Design

4.1 Why Comparisons Are Paired

The matcher LLM is not deterministic, and on these datasets its variation is large enough to swamp the effects being measured. Running the no-selection baseline twice on Fodors–Zagat, with identical prompts and identical attributes, produced F1 of 0.8917 and 0.9596 — a gap of 0.068, from 19 of 2,665 answers changing, 18 of them from Yes to No. Precision moved from 0.823 to 0.947 while recall was unchanged.

The same measurement on DBLP–ACM gives a much smaller floor: two independent draws of the no-selection baseline produced F1 of 0.8706 and 0.8613, a gap of 0.0093. This is expected — DBLP–ACM has 2,224 ground-truth matches against Fodors–Zagat’s 110, so the same absolute number of flipped answers perturbs the metrics far less.

Comparing two methods by their F1 values from separate runs therefore measures the matcher’s variance more than it measures the methods. Two consequences follow.

First, the matcher response cache is enabled for all comparisons. Where two methods select the same attributes for a pair, the pair yields one shared answer and cannot contribute a spurious difference. Section 5.4 shows this working: across the 9,803 pairs where two very different methods happened to select the same single attribute, they recorded identical false-positive counts.

Second, methods are compared by McNemar’s test on the pairs where they actually differ, rather than by comparing aggregate F1. This is a paired test on the same candidate set and the same matcher draw, and it is the only comparison in this report that carries a significance claim.

A caveat this design imposes: because repeated runs of one configuration replay the cache, they are not independent measurements and their standard deviation is meaningless. Replication across seeds was dropped in favour of the paired design.

4.2 Holding the Candidate Set Fixed Across Methods

Step 1.5 leaves blocking untouched, so its candidate set is identical to the no-selection baseline. Step 1 condenses the record and can optionally re-block on the condensed attributes, which would change which pairs are compared. It is therefore run with re-blocking disabled (`--step1-no-reblock`) throughout, so that every configuration is scored over the same 13,080 candidate pairs containing the same 2,224 ground-truth matches, and attribute choice is the only difference between them.

5 Results

5.1 Fodors–Zagat: A Saturated Benchmark

Every pairwise McNemar test returns $p = 1.00$. The largest disagreement between any two methods is 9 pairs out of 2,665, split almost evenly in both directions.

Method	TP	FP	FN	F1	Precision	Recall	Attrs/pair	Prompt tokens
Step 1.5, decisive	106	4	4	0.9636	0.9636	0.9636	2.68	652,422
Step 1.5, ordinal	108	6	2	0.9643	0.9474	0.9818	4.00	705,395
No selection (<code>full</code>)	107	6	3	0.9596	0.9469	0.9727	5.00	748,469
Step 1, supervised (10 labels)	106	5	4	0.9593	0.9550	0.9636	2.00	605,836

Table 1: Fodors–Zagat, `class` excluded. 2,665 candidate pairs, 110 ground-truth matches. Metrics are end-to-end. All rows use the cumulative rule at its 0.8 default.

This is a property of the benchmark, not of the methods. Doing nothing at all already reaches F1 0.9596 with 107 of 110 matches recovered and 6 false positives. There is no headroom in which a method can distinguish itself, and with 110 ground-truth pairs the matcher’s own instability moves F1 further than any method does.

What remains measurable is cost. The decisive formulation shows the matcher 2.68 attributes per pair instead of 5.00, a 46% reduction, changing 4 of 2,665 decisions, two in each direction. The defensible claim from this dataset is that adaptive selection is free: it does not cost accuracy, and it does not buy any.

5.2 DBLP–ACM: Where the Methods Separate

Method	Rule	Attrs/pair	TP	FP	FN	F1	Precision	Recall
Step 1.5, decisive	gap	1.42	2197	255	27	0.9397	0.8960	0.9879
Step 1, supervised (200 labels)	—	1.00	2198	339	26	0.9233	0.8664	0.9883
Step 1.5, decisive	cumulative 0.8	2.24	2214	535	10	0.8904	0.8054	0.9955
No selection (<code>full</code>)	—	4.00	2216	651	8	0.8706	0.7729	0.9964
Step 1.5, ordinal	cumulative 0.8	3.77	2215	699	9	0.8622	0.7601	0.9960
Step 1, supervised (10 labels)	—	2.00	2211	739	13	0.8547	0.7495	0.9942

Table 2: DBLP–ACM. 13,080 candidate pairs, 2,224 ground-truth matches. Metrics are end-to-end and every configuration is scored over the same candidate set (Section 4.2). Rows are ordered by F1.

Comparison (A vs. B)	A right	B right	p
Decisive/gap vs. no selection	527	150	5.2×10^{-50}
Decisive/gap vs. supervised (200 labels)	112	29	1.0×10^{-12}
Decisive/gap vs. decisive/cumulative 0.8	375	112	3.5×10^{-34}
Supervised (200) vs. decisive/cumulative 0.8	389	209	1.6×10^{-13}
Decisive/cum. 0.8 vs. no selection	468	350	4.2×10^{-5}
Decisive/cum. 0.8 vs. ordinal/cum. 0.8	425	258	1.7×10^{-10}
No selection vs. ordinal/cum. 0.8	135	86	1.2×10^{-3}
Ordinal/cum. 0.8 vs. supervised (10)	497	453	0.16

Table 3: McNemar tests over the pairs on which the two methods disagree, all on the identical 13,080-pair candidate set. Counts are pairs the named method got right and the other got wrong.

Three readings follow.

Recall is effectively constant; every difference is in precision. Across the table recall spans 0.988–0.996 while precision spans 0.750–0.896. Attribute selection does not help the matcher find matches it was missing — it stops the matcher from asserting matches that are not there. The one systematic cost is at the aggressive end: the configurations showing barely more than one attribute give up 18–19 of 2,224 matches relative to showing everything.

Elicitation decides whether adaptation helps at all. At the same selection rule, the decisive formulation beats no selection ($p = 4.2 \times 10^{-5}$) while the ordinal formulation is significantly *worse* than no selection ($p = 1.2 \times 10^{-3}$). The pipeline, sampler, transfer model, rule, and matcher are identical between them; only the profiling question differs. The ordinal formulation also barely selects, retaining all four attributes on 76.8% of pairs and averaging 3.77 of 4.

The attribute the method learns to discard is venue. The reason is visible in the data: the two sources render the same venue very differently — "SIGMOD Record" against "The VLDB Journal --- The International Journal on..." — so venue disagreement is evidence of formatting, not of non-identity. The ordinal formulation scores *venue* in isolation, where it looks moderately informative, and keeps it.

5.3 The Selection Rule Matters as Much as the Elicitation

The cumulative rule’s 0.8 target has been in place since the pipeline was written and had never been examined. Sweeping it downward, with the floor at one attribute and everything else held fixed, changes the outcome by more than any methodological variable in this report.

Rule	Attrs/pair	FP	FN	F1	title only
cumulative 0.8 (default)	2.24	535	10	0.8904	0.0%
cumulative 0.6	1.66	434	13	0.9082	34.5%
cumulative 0.5	1.27	293	20	0.9337	73.2%
cumulative 0.4	1.18	275	26	0.9359	82.7%
gap (no threshold)	1.42	255	27	0.9397	74.9%
supervised, fixed [title]	1.00	339	26	0.9233	100.0%

Table 4: DBLP–ACM, decisive profiling throughout, varying only how the ranked importance vector is turned into a subset. The last row is the labelled global baseline, shown as the limiting case of one attribute on every pair.

At the shipped default of 0.8, adaptive selection *loses* to the labelled global baseline ($p = 1.6 \times 10^{-13}$). At 0.4 it wins (66/2, $p = 1.6 \times 10^{-17}$). The comparison between adaptive and global selection therefore turned on an untuned hyperparameter rather than on the methods.

Two further observations. The best configuration is at neither extreme: showing everything (4.00 attributes) and showing one fixed attribute (1.00) are both beaten by rules that show roughly 1.2–1.4. And the cumulative targets 0.4 and 0.5 are statistically indistinguishable from each other (55/67, $p = 0.32$), so the effect is a plateau rather than a single lucky value, while the steps above them are individually significant.

The **gap** rule is the practically important row. It has no threshold, so nothing about it was chosen by looking at these results, and it reaches the top of the range anyway. Its advantage over the best hand-tuned target is thin (50/31, $p = 0.045$) and should not be claimed as superiority; the defensible claim is that it matches a tuned threshold without requiring one. Since it also does not saturate on wide schemas (Section 3.3), it is the rule used for the Amazon–Walmart runs now in progress.

5.4 Why Adaptation Beats a Fixed Subset

The labelled global baseline reduces every record to `title`, which is also what the adaptive method chooses for three quarters of pairs. Comparing the two therefore isolates adaptation itself: the same attribute vocabulary, applied uniformly against pair-by-pair.

Pair group	Pairs	FP, adaptive	FP, fixed [title]
Adaptive also chose [title]	9,803	118	118
Adaptive chose more	3,277	137	221
[title, authors, year]	2,257	101	146
[title, year]	841	13	58
[title, authors]	179	23	17

Table 5: DBLP-ACM, all 13,080 candidate pairs, comparing Step 1.5 decisive under the `gap` rule against the 200-label global baseline over the identical candidate set.

Where the two select the same attribute they produce identical results — 118 false positives each — which is what the shared matcher cache guarantees and which confirms the comparison is clean. The entire gain comes from the 3,277 pairs where the adaptive method kept more, cutting false positives from 221 to 137.

The clearest single effect is the 841 pairs given [title, year]: false positives fall from 58 to 13. These are pairs whose titles are similar enough to mislead a title-only matcher, and where the year settles the question. A global method cannot express this, because adding `year` to every pair is what the cumulative-0.8 configuration effectively does, and that costs more elsewhere than it saves here.

6 Discussion

Pair-adaptive selection outperforms both no selection and a labelled global baseline. On DBLP-ACM it improves F1 from 0.8706 to 0.9397 over showing the matcher everything ($p = 5.2 \times 10^{-50}$) while showing it 65% fewer attributes, and beats a baseline trained on 400 labelled pairs over the identical candidate set ($p = 1.0 \times 10^{-12}$) while using none.

How importance is elicited decides whether adaptation helps. The ordinal formulation, on the same pipeline and rule, is significantly worse than no selection at all. Asking the model to score each attribute in isolation produces a nearly flat importance vector that no selection rule can act on; asking it to name a sufficient subset forces the comparison between attributes that the task actually requires. This is the most transferable finding here, and it is a property of the elicitation, not of the dataset.

The rule that consumes the importance vector is as consequential as the vector. The same profiling output, read by the default rule, loses to the global baseline; read by `gap`, it beats it. Any claim about pair-adaptive selection that does not state its selection rule is under-specified. That the pipeline’s default was both untuned and, on wide schemas, structurally incapable of condensing at all is the most actionable finding in this report.

The benefit is conditional on the benchmark having headroom. On Fodors-Zagat, where the no-selection baseline already reaches F1 0.9596, no method differs from any other. DBLP-ACM has headroom, and it rewards adaptation even though `title` nearly identifies a paper on its own. That the margin over a fixed subset is nonetheless modest — 141 discordant pairs in 13,080

— is what a near-unique key predicts. A wide schema with no such key is the sharper test.

7 Limitations

Amazon–Walmart is missing. It is the widest schema available (13 attributes) and the least saturated baseline observed, so it is where adaptive selection should show its largest effect and where the near-unique-key caveat above does not apply. Runs are in progress; its absence is the main gap in the evidence.

The selection rule was examined on the evaluation set. The sweep in Section 5.3 was read on the same pairs it reports over, with no held-out split. It identifies a band worth selecting from, not a defensible value, and no single cumulative target from it should be quoted. This is why **gap** is preferred: having no threshold, it is not exposed to this objection, though it was still chosen after seeing these results rather than before.

Single matcher draw. Each configuration was run once. The paired design makes the comparisons valid within that draw — differing pairs are counted directly, agreeing pairs share one answer — but it does not establish that the effect sizes replicate exactly under a different draw. The large DBLP–ACM effects, with hundreds of discordant pairs, are unlikely to reverse; the **gap**-versus-tuned-threshold comparison, with 81 discordant pairs and $p = 0.045$, could.

The supervised baseline differs in supervision, not only in method. It uses 400 labelled pairs; the adaptive configurations use none. Comparing them measures method and label budget together. Its selected subset also moves with the label budget — `[title, authors]` at 10 and 50 labels, `[title]` at 200 — so no single sample size characterises it.

The feature vector is weaker than it appears. Of the six per-attribute features the transfer model consumes, `semantic_similarity` is disabled and constant at zero, and `is_numeric` is constant per column on both datasets reported here. On the widest schema `exact_match` fires on 0.06% of cells. Effectively three features carry signal. The vector also has no notion of value rarity, which is the standard discriminative signal in entity resolution: matching on a rare title token and on a common one are represented identically.

One sampler. All Step 1.5 runs use the `diversity` profile sampler. No claim about sampler choice is made here.

8 Remaining Work

1. Complete the Amazon–Walmart runs and repeat this comparison on a wide schema with no near-unique key. This is the decisive test of whether the result generalizes.
2. Select the selection rule and any threshold on held-out pairs rather than on the evaluation set. **gap** removes the threshold but not the rule choice.
3. Repeat the DBLP–ACM comparison under an independent matcher draw, to confirm the effect sizes.
4. Add a value-rarity feature and re-measure; the current feature set gives the transfer model no way to distinguish a distinctive match from a generic one.
5. Re-establish the sampler comparison on the corrected pipeline, on DBLP–ACM rather than Fodors–Zagat, where the effect can be resolved.

9 Summary

On DBLP–ACM, pair-adaptive attribute selection driven by a subset-elicitation prompt and the `gap` selection rule reaches F1 0.9397 using 1.42 of 4 attributes per pair. It beats showing the matcher every attribute (0.8706, $p = 5.2 \times 10^{-50}$) and beats a global baseline trained on 400 labelled pairs that reduces every record to `title` (0.9233, $p = 1.0 \times 10^{-12}$), while using no labels itself. All gains are in precision; recall falls slightly at the aggressive end. On Fodors–Zagat, where the no-selection baseline already reaches F1 0.9596, no method is distinguishable from any other, and the defensible claim is a 46% reduction in attributes at unchanged accuracy.

Two conditions govern the result. The LLM must be asked for a decision rather than for a score: the same pipeline driven by an ordinal scoring prompt is significantly worse than no selection at all. And the rule turning importance into a subset must be able to express small subsets: under the pipeline’s own default the same method loses to the global baseline it otherwise beats. Neither condition is visible from the aggregate metric alone, which is why both are reported here.